

CASE STUDY

Migrating a Live HPC Design Environment to a Resilient, Scalable Infrastructure

How Opt IT Technologies helped a leading semiconductor organization eliminate instability, reduce costs, and build a future-ready compute foundation.



Challenge

The customer's High-Performance Computing (HPC) design environment was running on a virtual setup that struggled to scale across multiple simultaneous programs.

Instability, lack of redundancy, and limited high-availability options were impacting the productivity of design teams dependent on continuous access to compute, storage, and display resources.

Domain
**Semiconductor EDA /
HPC Infrastructure**



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Solution

Infrastructure Migration

Opt IT Technologies designed and executed a migration to a consolidated, high-performance infrastructure spanning compute, display, and storage/backup across two datacenters running in parallel.

Planned Execution

The project was managed with rigorous planning to ensure zero disruption to live design workflows. Regular updates were communicated to both end users and management throughout the execution.

Infrastructure Optimization

Opt IT leveraged its expertise to successfully migrate the environment, eliminate redundant licenses, and optimize existing hardware for improved performance and reliability.



Expertise Used

- Semiconductor Design Stack
- Compute Architecture
- Storage
- Virtualization
- EDA Workflows
- LSF Workload Management

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KEY OUTCOMES

ZERO DOWNTIME

Live environment migrated with no disruption to design users.

COST REDUCTION

Unnecessary licenses identified and eliminated.

HIGH AVAILABILITY

Redundancy built into compute, storage, and display layers.

DUAL-DATACENTER

Parallel setup ensures resilience and future scalability.

CUSTOMER TESTIMONIAL

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We initially set up a High-Performance Computing Design environment on a virtual environment that had to scale to manage multiple programs simultaneously.

Opt IT Technologies helped us migrate to a stable, high-performance computing environment by consolidating our display, compute, and storage/backup systems into a well-architected infrastructure.

Based on our experience working with the Opt IT team, I would like to highly recommend them as a strong IT infrastructure partner for any organization trying to set up a design environment built for scale. They are very professional and provide quality support.

~ *Head, Leading Semiconductor Organization*



Contact Us



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